

VED NARVEKAR

BACKEND & AI ENGINEER

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SUMMARY

Backend & AI Engineer with hands-on experience building production-grade systems — high-throughput API gateways handling **~4,000 RPS**, autonomous LLM-powered CLI agents with multi-provider support, and ML-backed web applications with measurable performance outcomes. Proficient in Node.js, TypeScript, Python, system design, and applied AI/ML. Seeking backend or AI engineering internships to ship reliable, impactful production features.

EDUCATION

B.E. Artificial Intelligence & Machine Learning — Mumbai University

Expected Apr 2028

Universal College of Engineering, Mumbai | Discipline & Anti-Ragging Committee Head

EXPERIENCE

Backend Engineer Intern — SR India

Jan 2026 – Feb 2026 · Remote

- Engineered **REST APIs** for a hotel booking platform (Node.js, Express, PostgreSQL) covering **4 core modules** — bookings, room availability, user management, and admin controls — with **JWT auth and RBAC**.
- Improved API performance and reliability through targeted **query optimization**, input validation, and **Redis caching**; eliminated redundant DB calls across high-frequency booking endpoints.
- Integrated all endpoints into a React dashboard and maintained **Postman collections** documenting request/response contracts, enabling zero back-and-forth handoffs with the frontend team.

PROJECTS

resolv — AI-Powered GitHub Issue Resolver CLI

github.com/vednarvekar/resolv-v2

TypeScript · Node.js · LLM APIs · RAG Pipelines · ts-morph

- Autonomous CLI agent that fetches GitHub issues, fingerprints repo coding patterns via **RAG** (ts-morph DNA extraction + NIM embeddings), and generates **style-matched patches** via an LLM tool-calling loop in **under 60 seconds**.
- Unified **7 LLM providers** (Anthropic, OpenAI, Gemini, Grok, OpenRouter, NVIDIA NIM, Ollama) behind a single OpenAI-compatible interface — provider switch requires one config change.
- Shipped web search (Brave + DuckDuckGo fallback), codebase grep/write tools, persistent chat sessions, and repo-context compression cutting **LLM token usage**.

API Gateway — High-Performance Reverse Proxy

github.com/vednarvekar/api-gateway

Node.js · Fastify · TypeScript · Redis · PostgreSQL · Prometheus

- Architected a production-grade API gateway from scratch reaching **~4,000 RPS at 250 concurrent users**, outperforming naive Express proxies by **4×**; load-tested with autocannon.
- Implemented **Redis sliding-window rate limiting**, circuit breaking (opossum), response caching, JWT/RBAC, Prometheus metrics, and Drizzle ORM migrations, dynamic routing with PostgreSQL.

Pixel — AI Media Detection

github.com/vednarvekar/pixel-v2

React · TypeScript · FastAPI · PyTorch · PostgreSQL · Supabase

- Built a full-stack image and video scanner that fuses **ResNet-18 inference**, EXIF/XMP/IPTC and C2PA metadata heuristics, and **Google Vision Web Detection** into a weighted AI-probability score (default **85% model / 10% metadata / 5% web**), verdict, reasoning, and signal breakdown.
- Orchestrated React/Vite, Express/TypeScript, and a FastAPI/PyTorch model service with PostgreSQL scan history and Supabase email/Google OAuth; supports media up to **10 MB images** and **100 MB videos** with 30-second video analysis.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, C++
Backend	Node.js, Fastify, Express.js, REST APIs, System Design, Authentication, RBAC
AI / ML	PyTorch, NumPy, Pandas, Scikit-learn, LangChain, RAG Pipelines, LLM APIs
Databases	PostgreSQL, MongoDB, Redis, Drizzle ORM, Database Design
DevOps	Linux, Git, GitHub Actions, Prometheus, Docker, AWS, Postman
Frontend	React, Next.js, HTML, CSS, MERN Stack

CERTIFICATIONS

- Complete Data Science, Machine Learning, DL & NLP Bootcamp
- Node.js, Express, MongoDB & More: The Complete Bootcamp
- The Ultimate React Course 2025: React, Next.js, Redux & More

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